



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

Introduction to Python for Data Science

DSECLPFDS / AIMLCPFDS

- Immutable & Mutable Data Structures
- Lists
- Dictionaries
- Sets
- Conditional Statements

Agenda for CS#3



- Exception Handling
- Functions
- Files
- SciPy Ecosystem

- Even if a statement or expression is syntactically correct, it may cause an error when an attempt is made to execute it. Errors detected during execution are called *exceptions* and are not unconditionally fatal

- Allow us to package functionality in a nice and readable way
- reuse it without writing it again
- Make code modular and readable
- Rule of thumb - if you are planning on using very similar code more than once, it may be worthwhile writing it as a reusable function.

Function Declaration



keyword

Any number of arguments

```
def functionName(argument1, argument2, argument3, ... argumentN):  
    statements..
```

```
    ..  
    ..
```

```
    return returnValue
```

[Optional] Exits the function and returns some value

Built-in Functions



Built-in Functions				
<code>abs()</code>	<code>dict()</code>	<code>help()</code>	<code>min()</code>	<code>setattr()</code>
<code>all()</code>	<code>dir()</code>	<code>hex()</code>	<code>next()</code>	<code>slice()</code>
<code>any()</code>	<code>divmod()</code>	<code>id()</code>	<code>object()</code>	<code>sorted()</code>
<code>ascii()</code>	<code>enumerate()</code>	<code>input()</code>	<code>oct()</code>	<code>staticmethod()</code>
<code>bin()</code>	<code>eval()</code>	<code>int()</code>	<code>open()</code>	<code>str()</code>
<code>bool()</code>	<code>exec()</code>	<code>isinstance()</code>	<code>ord()</code>	<code>sum()</code>
<code>bytearray()</code>	<code>filter()</code>	<code>issubclass()</code>	<code>pow()</code>	<code>super()</code>
<code>bytes()</code>	<code>float()</code>	<code>iter()</code>	<code>print()</code>	<code>tuple()</code>
<code>callable()</code>	<code>format()</code>	<code>len()</code>	<code>property()</code>	<code>type()</code>
<code>chr()</code>	<code>frozenset()</code>	<code>list()</code>	<code>range()</code>	<code>vars()</code>
<code>classmethod()</code>	<code>getattr()</code>	<code>locals()</code>	<code>repr()</code>	<code>zip()</code>
<code>compile()</code>	<code>globals()</code>	<code>map()</code>	<code>reversed()</code>	<code>__import__()</code>
<code>complex()</code>	<code>hasattr()</code>	<code>max()</code>	<code>round()</code>	
<code>delattr()</code>	<code>hash()</code>	<code>memoryview()</code>	<code>set()</code>	

To find out how they work: <https://docs.python.org/3.3/library/functions.html>

- Opening and closing files
 - `the_file = open(filename, mode)`
 - `the_file.close()`
- Reading files
 - `string = the_file.read(number_of_characters)`
 - `string = the_file.readline(number_of_characters)`
 - `list_of_strings = the_file.readlines()`
- Writing files
 - `the_file.write(string)`
 - `the_file.writelines(list_of_strings)`

Mode	Description
"r"	Read from a file. If the file doesn't exist, Python will complain with an error.
"w"	Write to a file. If the file exists, its contents are overwritten. If the file doesn't exist, it's created.
"a"	Append a file. If the file exists, new data is appended to it. If the file doesn't exist, it's created.
"r+"	Read from and write to a file. If the file doesn't exist, Python will complain with an error.
"w+"	Write to and read from a file. If the file exists, its contents are overwritten. If the file doesn't exist, it's created.
"a+"	Append and read from a file. If the file exists, new data is appended to it. If the file doesn't exist, it's created.

Introduction to SciPy Ecosystem

SciPy (pronounced “Sigh Pie”) is a Python-based ecosystem of open-source software for mathematics, science, and engineering. In particular, these are some of the core packages:



NumPy
Base N-dimensional
array package



SciPy library
Fundamental library for
scientific computing



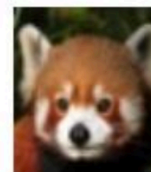
Matplotlib
Comprehensive 2-D
plotting



IPython
Enhanced interactive
console



SymPy
Symbolic mathematics



pandas
Data structures &
analysis

<https://www.scipy.org/>
<https://www.scipy.org/about.html>

Numpy

NumPy stands for Numerical Python.

- NumPy is a Python library used for working with arrays.
- It also has functions for working in domain of linear algebra, fourier transform, and matrices.

Why use NumPy ?

- In Python we have lists that serve the purpose of arrays, but they are slow to process.
- NumPy aims to provide an array object that is up to 50x faster than traditional Python lists.
- The array object in NumPy is called *ndarray*, it provides a lot of supporting functions.